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Ohori; Ryo et al.

Motor control device, motor device, wiper device, and motor control method

Abstract

A motor control device which controls a motor having an output shaft includes a vehicle speed detection part and a driving control part. The vehicle speed detection part detects a vehicle speed, which is a traveling speed of a vehicle mounted with the motor. Based on the vehicle speed detected by the vehicle speed detection part, the driving control part performs control of switching between a first driving mode which drives the motor, and a second driving mode in which a number of revolutions of the output shaft and an output of the motor are higher than in the first driving mode.

Inventors: Ohori; Ryo (Gunma, JP), Amagasa; Toshiyuki (Gunma, JP), Yamada; Masahiro (Gunma, JP), Mochida; Atsushi (Gunma, JP), Iwazaki; Tamotsu (Gunma, JP)

Applicant: MITSUBA Corporation (Gunma, JP)

Family ID: 90140163

Assignee: MITSUBA Corporation (Gunma, JP)

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Primary Examiner: Agared; Gabriel

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Japan application serial no. 2022-153220, filed on Sep. 27, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a motor control device, a motor device, a wiper device, and a motor control method.

Related Art

(3) In recent years, in motor control, motor control devices that switch between different driving modes, such as a non-free driving mode and a rectangular wave driving mode, according to a load magnitude have been known (e.g., see Patent Document 1: Japanese Patent Application Laid-Open No. 2020-48401). Herein, the non-free driving mode is a driving mode that outputs one half of the duty in the OFF phase (corresponding to a phase open period, hereinafter also referred to as a free phase) among three phases, and the non-free driving mode is a low-output driving mode in which the number of revolutions decreases compared to the rectangular wave driving mode in the case of

driving at the same power. In Patent Document 1, the switching control on the driving mode is performed based on a load integrated value, which is a value determined according to a load and is obtained by integrating the load.

(4) In the driving of a motor mounted on a vehicle, such as a wiper device, a high load may be generated from an initial stage of driving, for example, during vehicle traveling. However, in the motor control device as described above, in the case where a high load is generated from an initial stage of driving, since the load has not yet been integrated, the driving mode may be switched to a low-output driving mode such as the non-free driving mode, and there is a possibility that motor driving may stop halfway.

SUMMARY

(5) An aspect of the disclosure is a motor control device which controls a motor having an output shaft. The motor control device includes a vehicle speed detection part and a driving control part. The vehicle speed detection part detects a vehicle speed, which is a traveling speed of a vehicle mounted with the motor. Based on the vehicle speed detected by the vehicle speed detection part, the driving control part performs control of switching between a first driving mode which drives the motor, and a second driving mode in which a number of revolutions of the output shaft and an output of the motor are higher than in the first driving mode.

(6) Further, an aspect of the disclosure is a motor control method which controls a motor having an output shaft. The motor control method includes a vehicle speed detection step and a driving control step. In the vehicle speed detection step, a vehicle speed detection part detects a vehicle speed, which is a traveling speed of a vehicle mounted with the motor. In the driving control step, based on the vehicle speed detected in the vehicle speed detection step, a driving control part performs control of switching between a first driving mode which drives the motor, and a second driving mode in which a number of revolutions of the output shaft and an output of the motor are higher than in the first driving mode.

(7) According to the embodiments of the disclosure, it is possible to appropriately perform motor driving even in the case where a high load is generated from an initial stage of motor driving.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram showing an example of a motor device according to this embodiment.

(2) FIG. 2 is a diagram showing an example of rectangular wave driving patterns in this embodiment.

(3) FIG. 3 is a table summarizing an example of the rectangular wave driving patterns in this embodiment.

(4) FIG. 4 is a diagram showing a first example of non-free driving patterns in this embodiment.

(5) FIG. 5 is a table summarizing the first example of the non-free driving patterns in this embodiment.

(6) FIG. 6 is a diagram showing a second example of the non-free driving patterns in this embodiment.

(7) FIG. 7 is a table summarizing the second example of the non-free driving patterns in this embodiment.

(8) FIG. 8 is a diagram showing an example of a switching operation of a driving mode of a motor control device according to this embodiment.

(9) FIG. 9 is a flowchart showing an example of the switching operation of the driving mode of the motor control device according to this embodiment.

(10) FIG. 10 is a flowchart showing another example of the switching operation of the driving mode of the motor control device according to this embodiment.

(11) FIG. 11 is a flowchart showing a modification example of the switching operation of the driving mode of the motor control device according to this embodiment.

(12) FIG. 12 is a configuration view showing an example of a wiper device according to this embodiment.

DESCRIPTION OF THE EMBODIMENTS

(13) Embodiments of the disclosure provide a motor control device, a motor device, a wiper device, and a motor control method capable of appropriately performing motor driving even in the case where a high load is generated from an initial stage of motor driving.

(14) Hereinafter, a motor control device, a motor device, a wiper device, and a motor control method according to an embodiment of the disclosure will be described with reference to the drawings.

(15) FIG. 1 is a block diagram showing an example of a motor device **100** according to this embodiment. As shown in FIG. 1, the motor device **100** includes a motor **2**, a control part **40**, and an inverter **50**. The motor device **100** according to this embodiment is, for example, used in a wiper device that wipes a window glass of a vehicle.

(16) In this embodiment, the control part **40** and the inverter **50** are compatible with a motor control device **150**. Further, a battery **3** and an engine control unit (ECU) **4** are connected to the motor control device **150**.

(17) The motor **2** is, for example, a three-phase four-pole brushless motor. The motor **2** rotates and drives according to an output signal (applied voltage) outputted by the inverter **50** based on a driving signal to be described later. Further, the motor **2** includes a stator **21** and a rotor **22**.

(18) The stator **21** is fixed to an inner circumference of a case of the motor **2**. The stator **21** includes windings (**21u**, **21v**, **21w**) of three phases. The stator **21** is wound with the windings (**21u**, **21v**, **21w**). For example, the windings (**21u**, **21v**, **21w**) of three phases are connected in a delta connection configuration.

(19) In the delta connection configuration, the winding **21u** and the winding **21w** are connected by a connection point **21a**, the winding **21v** and the winding **21w** are connected by a connection point **21c**, and the winding **21u** and the winding **21v** are connected by a connection point **21b**.

(20) The rotor **22** is provided on the inner side of the stator **21**. The rotor **22** includes, for example, a rotor shaft **22a** and a four-pole permanent magnet **22b** attached to the rotor shaft **22a**. A plurality of bearings (not shown) are provided in the case of the motor **2**, and the rotor shaft **22a** is rotatably supported by the plurality of bearings.

(21) A position detection part **30** detects a signal corresponding to rotation of the rotor **22**. The position detection part **30** includes, for example, three Hall elements (**30u**, **30v**, **30w**). Upon rotation of the rotor **22**, the three Hall elements (**30u**, **30v**, **30w**) output pulse signals, which are 120 degrees phase-shifted with respect to each other, to the control part **40**. That is, the position detection part **30** generates pulse signals based on changes in the magnetic pole of a sensor magnet (not shown) arranged at the rotor shaft **22a** as the rotor **22** rotates, and outputs the pulse signals to the control part **40**. Each of the Hall element (**30u**, **30v**, **30w**) detects a position shifted by 120 degrees in electrical angle.

(22) In this embodiment, the Hall element **30u** outputs a digital signal (position detection signal Hu) corresponding to the U-phase, the Hall element **30v** outputs a digital signal (position detection signal Hv) corresponding to the V-phase, and the Hall element **30w** outputs a digital signal (position detection signal Hw) corresponding to the W-phase. In this embodiment, the three Hall elements (**30u**, **30v**, **30w**) are arranged with respect to the rotor **22** in such a manner that an advance angle of 30 degrees in electrical angle is obtained in the case where the output of the inverter **50** is changed immediately at each position at which the level of the output signal of the three Hall elements (**30u**, **30v**, **30w**) changes, i.e., at each position at which an edge occurs in the output signal.

(23) Based on a driving signal generated by the driving control part **43** (to be described later), the

inverter **50** is subjected to, for example, pulse width modulation (PWM) control and applies a voltage to the windings (**21u**, **21v**, **21w**) of three phases of the motor **2**. That is, based on the driving signal generated by the driving control part **43**, the inverter **50** causes switching elements (**51a** to **51f**) to perform a switching operation (conduction/non-conduction) to change an output magnitude (duty ratio), an energization period (energization angle), and an energization timing (advance angle) of the voltage applied to the motor **2**. Herein, the duty ratio represents a ratio of a conduction period of the corresponding switching element in a PWM cycle.

(24) The inverter **50** generates an applied voltage with a direct current power supplied from the battery **3**. The battery **3** is, for example, a direct current power supply such as a lead-acid battery or a lithium-ion battery and supplies power that drives the motor **2**.

(25) The inverter **50** includes six switching elements **51a** to **51f** connected in a three-phase bridge, and diodes **52a** to **52f**. The switching elements **51a** to **51f** are, for example, N-channel metal oxide semiconductor field effect transistors (MOSFET) and constitute a three-phase bridge circuit.

(26) The switching element **51a** and the switching element **51d** are connected in series between a positive electrode terminal and a negative electrode terminal of the battery **3** and constitute a U-phase bridge circuit. The switching element **51a** has a drain terminal connected to the positive electrode terminal of the battery **3**, a source terminal connected to a node N1, and a gate terminal connected to a signal line of an upper-side driving signal of the U-phase. Further, the switching element **51d** has a drain terminal connected to the node N1, a source terminal connected to the negative electrode terminal of the battery **3**, and a gate terminal connected to a signal line of a lower-side driving signal of the U-phase. Further, the node N1 is connected to the connection point **21a** of the motor **2**.

(27) The switching element **51b** and the switching element **51e** are connected in series between the positive electrode terminal and the negative electrode terminal of the battery **3** and constitute a V-phase bridge circuit. The switching element **51b** has a drain terminal connected to the positive electrode terminal of the battery **3**, a source terminal connected to a node N2, and a gate terminal connected to a signal line of an upper-side driving signal of the V-phase. Further, the switching element **51e** has a drain terminal connected to the node N2, a source terminal connected to the negative electrode terminal of the battery **3**, and a gate terminal connected to a signal line of a lower-side driving signal of the V-phase. Further, the node N2 is connected to the connection point **21b** of the motor **2**.

(28) The switching element **51c** and the switching element **51f** are connected in series between the positive electrode terminal and the negative electrode terminal of the battery **3** and constitute a W-phase bridge circuit. The switching element **51c** has a drain terminal connected to the positive electrode terminal of the battery **3**, a source terminal connected to a node N3, and a gate terminal connected to a signal line of an upper-side driving signal of the W-phase. Further, the switching element **51f** has a drain terminal connected to the node N3, a source terminal connected to the negative electrode terminal of the battery **3**, and a gate terminal connected to a signal line of a lower-side driving signal of the W-phase. Further, the node N3 is connected to the connection point **21c** of the motor **2**.

(29) Further, the diode **52a** has an anode terminal connected to the node N1 and a cathode terminal connected to the positive electrode terminal of the battery **3**. Further, the diode **52d** has an anode terminal connected to the negative electrode terminal of the battery **3** and a cathode terminal connected to the node N1.

(30) Further, the diode **52b** has an anode terminal connected to the node N2 and a cathode terminal connected to the positive electrode terminal of the battery **3**. Further, the diode **52e** has an anode terminal connected to the negative electrode terminal of the battery **3** and a cathode terminal connected to the node N2.

(31) Further, the diode **52c** has an anode terminal connected to the node N3 and a cathode terminal connected to the positive electrode terminal of the battery **3**. Further, the diode **52f** has an anode

terminal connected to the negative electrode terminal of the battery 3 and a cathode terminal connected to the node N3.

(32) The control part 40 is, for example, a processor including a central processing unit (CPU) and comprehensively controls the motor device 100. The control part 40 generates a driving signal corresponding to a target rotation output (e.g., target number of revolutions TRPM) of the rotor 22 and outputs the generated driving signal to the inverter 50. The control part 40 transmits and receives predetermined information to and from the ECU 4 and the motor 2. Further, the control part 40 includes a load integrated value generation part 41, a vehicle speed detection part 42, and a driving control part 43.

(33) The load integrated value generation part 41 executes a high load detection process of the rotor 22 based on a set rotational speed (number of revolutions) of the rotor 22 and a duty ratio. The load integrated value generation part 41 calculates and accumulates a load point value from the motor speed (rotational speed of the rotor 22) and the duty ratio, and generates an accumulated point value (load integrated value). The rotational speed of the rotor 22 is detected based on the output signals of the three Hall elements (30u, 30v, 30w).

(34) For example, referring to a load point map (not shown) that associates a motor speed, a duty ratio, and a power supply voltage with a load point value, the load integrated value generation part 41 acquires the load point value corresponding to the motor speed, the duty ratio, and the power supply voltage. The load integrated value generation part 41 integrates the acquired load point values and generates an accumulated point value as a load integrated value.

(35) As a high load state continues, since the “+” (positive) load point value continues, the accumulated point value (load integrated value) becomes a large positive value. On the other hand, as a normal load or a light load state continues, since the “0” or “-” (negative) load point value continues, the accumulated point value becomes “0” or less. Herein, the accumulated point value is all set to “0” in the case of “0” or less, and the accumulated point value indicates “0” when the motor 2 is operating normally. Further, once in the high load state, in the case where the load is subsequently reduced to a controllable region, the accumulated load point value is gradually subtracted and eventually converges to “0” or a small positive value. Thus, by looking at the accumulated point value, it is possible to learn the current status of the motor 2 and determine it as a high load in the case where the value exceeds a specific level. In this manner, the accumulated point value (load integrated value) serves as an index value of the load on the motor 2, and the control part 40 uses the accumulated point value (load integrated value) in the control on motor driving.

(36) The vehicle speed detection part 42 detects a vehicle speed, which is a traveling speed of the vehicle mounted with the motor 2. The vehicle speed detection part 42 detects the vehicle speed based on a vehicle speed signal outputted from the ECU 4.

(37) The driving control part 43 generates a driving signal corresponding to a target rotation output (e.g., target number of revolutions TRPM) of the rotor 22 and outputs the generated driving signal to the inverter 50. The driving control part 43, for example, switches between a rectangular wave driving mode (an example of a second driving mode) and a non-free driving mode (an example of a first driving mode) to control driving of the motor 2. The rectangular wave driving mode is a high-output driving mode in which a minimum rotational speed at which the rotor shaft 22a can be rotationally driven by the inputted power is higher than that in the non-free driving mode. Herein, referring to FIG. 2 to FIG. 7, details of the rectangular wave driving mode and the non-free driving mode will be described.

(38) FIG. 2, FIG. 4, and FIG. 6 are diagrams for illustrating an example of the position detection signals (Hu, Hv, Hw) outputted by the Hall elements (30u, 30v, 30w) and advance angles and energization angles in energization control of the inverter 50. FIG. 2, FIG. 4, and FIG. 6 show the correspondence between the position detection signals (Hu, Hv, Hw) and angular regions in which the switching elements 51a to 51f are turned on. The horizontal axis represents the rotational

position of the magnetic pole of the rotor 22 of the motor 2 in electrical angle.

(39) The position detection signals (Hu, Hv, Hw) have a phase difference of 120 degrees from each other with an electrical angle of 360 degrees being one cycle, and change to an H state (high state) or an L state (low state) every 180 degrees. In this embodiment, the change of the position detection signal Hu from the L state to the H state is referred to as a Hall edge HE1, and the change from the H state to the L state is referred to as a Hall edge HE4. Further, the change of the position detection signal Hv from the L state to the H state is referred to as a Hall edge HE3, and the change from the H state to the L state is referred to as a Hall edge HE6. Further, the change of the position detection signal Hw from the L state to the H state is referred to as a Hall edge HE5, and the change from the H state to the L state is referred to as a Hall edge HE2.

(40) Assuming that the position detection signals (Hu, Hv, Hw) outputted by the Hall elements (30u, 30v, 30w) do not contain errors, the electrical angle between the respective Hall edges is 60 degrees. Further, the region between the Hall edge HE1 and the Hall edge HE2 is referred to as a Hall stage 1 (hereinafter simply referred to as a stage 1 (the same applies hereinafter)), the region between the Hall edge HE2 and the Hall edge HE3 is referred to as a stage 2, and the region between the Hall edge HE3 and the Hall edge HE4 is referred to as a stage 3. Further, the region between the Hall edge HE4 and the Hall edge HE5 is referred to as a stage 4, the region between the Hall edge HE5 and the Hall edge HE6 is referred to as a stage 5, and the region between the Hall edge HE6 and the Hall edge HE1 is referred to as a stage 6.

(41) FIG. 2 is a diagram showing an example of rectangular wave driving patterns in this embodiment. FIG. 2 is a diagram showing the correspondence between the position detection signals (Hu, Hv, Hw) and energization patterns of the respective switching element 51a to 51f, with the horizontal axis representing the electrical angle. The example of energization control shown in FIG. 2 is the case where the advance angle is 20 degrees and the energization angle is 130 degrees.

(42) The energization pattern is a combination of any of states of the respective switching element 51a to 51f, including a state of being continuously turned on ("ON"), a state of being continuously turned off ("OFF") (period other than "ON" or "PWM", also referred to as a free phase period), and a state of being controlled to be turned on or off at a specific cycle (PWM-controlled state) ("PWM"). Each of the stages 1 to 6 is further divided into three sections including a section A', a section B', and a section C'. An energization pattern is set individually for each of the section A', the section B', and the section C'. The period (electrical angle) of each of the section A', the section B', and the section C' changes according to the value of the advance angle and the value of the energization angle.

(43) For example, in the stage 1 defined between the Hall edge HE1 and the Hall edge HE2, the energization pattern of the section A' is a combination of the states of the switching elements 51a to 51f being "ON", "OFF", "PWM", "PWM", "OFF", and "OFF", respectively. Further, the energization pattern of the section B' is a combination of the states of the switching elements 51a to 51f being "ON", "OFF", "PWM", "PWM", "ON", and "OFF", respectively. Further, the energization pattern of the section C' is a combination of the states of the switching elements 51a to 51f being "OFF", "OFF", "PWM", "PWM", "ON", and "OFF", respectively.

(44) Further, FIG. 3 is a table summarizing an example of the rectangular wave energization patterns in this embodiment shown in FIG. 2. A ROM (not shown) in the control part 40, for example, stores the rectangular wave energization patterns in a format as shown in FIG. 3. In FIG. 3, "1" represents "ON", "0" represents "OFF", and "P" represents "PWM".

(45) As described above, the combination defining the rectangular wave energization pattern includes three states (1) to (3) below.

(46) (1) First state: The switching element 51a to 51f is continuously turned on ("ON").

(47) (2) Second state: The switching element 51a to 51f is continuously turned off ("OFF") (period other than "ON" or "PWM").

(48) (3) Third state: The switching element 51a to 51f is controlled to be turned on or off at a

specific cycle (PWM-controlled state) (“PWM”).

(49) Thus, in the rectangular wave driving mode, the driving control part **43** controls conduction (energization) of the switching elements **51a** to **51f** according to the rectangular wave energization patterns above.

(50) Next, referring to FIG. **4** to FIG. **7**, the non-free driving mode will be described. There are two methods for controlling a non-free energization pattern used in the non-free driving mode.

(51) In a first method of non-free energization, a duty ratio of a PWM signal for the switching element connected to a coil of one phase among the three phases is set to an intermediate value between duty ratios of respective PWM signals for the switching elements connected to coils of the other two phases.

(52) In a second method of non-free energization, the duty ratio of the PWM signal for the switching element connected to the coil of one phase among the three phases is set to one half of an instruction duty ratio inputted from outside, and the duty ratios of the respective PWM signals for the switching elements connected to the coils of the other two phases are respectively set to the same value as the instruction duty ratio and 100%.

(53) <First Method of Non-Free Energization>

(54) In the driving and energization based on the non-free energization pattern using the first method, the states (1) to (3) above are respectively changed to three states (4) to (6) below.

(55) (4) Fourth state: The first state is changed to a fourth state (hereinafter referred to as a “PL” state) in which PWM control is performed according to a PWM signal with a maximum duty ratio greater than that in the PWM control in the third state.

(56) (5) Fifth state: The third state is changed to a fifth state (hereinafter referred to as a “PS” state) in which PWM control is performed according to a PWM signal with a minimum duty ratio smaller than that in the PWM control in the third state.

(57) (6) Sixth state: The second state is changed to a sixth state (hereinafter referred to as a “PM” state) in which PWM control is performed according to a PWM signal with an intermediate duty ratio between the maximum duty ratio and the minimum duty ratio.

(58) In other words, the timings at which the U-phase, V-phase, and W-phase windings (**21u**, **21v**, **21w**) respectively become the PM state (sixth state) in the non-free energization pattern are the same as the timings at which the U-phase, V-phase, and W-phase windings (**21u**, **21v**, **21w**) respectively become OFF-phase coils (second state) in the rectangular wave energization pattern.

(59) Thus, it is possible to address the problem that the control circuit (driving control part **43**) that drives the switching elements **51a** to **51f** malfunctions when a negative voltage is generated at an input terminal of the motor **2** upon switching of the energization pattern from the first state to the second state. Further, even at the timing (phase open period: period of being in the second state) of being free in 120-degree rectangular wave energization, PWM control is performed as in the sixth state, which results in 180-degree energization and a smooth current waveform at the time of commutation, so it becomes possible to expect an effect of quieting a driving sound (reducing motor operation sound).

(60) Herein, in this embodiment, the intermediate duty ratio is 50%. Further, the maximum duty ratio is a duty ratio obtained by adding one half of the instruction duty ratio inputted from outside to the intermediate duty ratio. Further, the minimum duty ratio is a duty ratio obtained by subtracting one half of the instruction duty ratio from the intermediate duty ratio.

(61) For example, in the case where the instruction duty ratio is 80%, since the intermediate duty ratio is preset to 50%, the maximum duty ratio becomes 90% according to $(50+80\div 2)$, and the minimum duty ratio becomes 10% according to $(50-80\div 2)$. The instruction duty ratio is assumed to be stored in advance to the ROM (not shown) of the control part **40** by a user.

(62) Herein, the switching elements **51d** to **51f** on the negative electrode side receive PWM signals that are in reverse phase to the PWM signals inputted to the switching elements **51a** to **51c** on the positive electrode side. Thus, the duty ratios of PWM signals that drive a pair of switching

elements on the positive electrode side and the negative electrode side differ from each other. However, in this embodiment, the duty ratios of PWM signals that drive the switching elements **51a** to **51c** on the positive electrode side are referred to as the duty ratios of PWM signals that drive the pairs of switching elements.

(63) FIG. 4 is a diagram showing an example of non-free energization patterns of the first method. FIG. 4 is a diagram showing the correspondence between the position detection signals (Hu, Hv, Hw) and the energization patterns of the respective switching elements **51a** to **51f**, with the horizontal axis representing the electrical angle.

(64) The example of energization control shown in FIG. 4 is the case where the advance angle is 20 degrees and the energization angle is 130 degrees. The energization pattern is a combination of any of states of the respective switching element **51a** to **51f**, including a state changed from being continuously turned on ("ON"), i.e., "ON" (first state), to "PL" (fourth state), a state changed from being continuously turned off ("OFF") (period other than "ON" or "PWM"), i.e., "OFF" (second state), to "PM" (sixth state), and a state changed from being controlled to be turned on or off at a specific cycle (PWM-controlled state) ("PWM"), i.e., "PWM" (third state), to "PS" (fifth state).

(65) Each of the stages 1 to 6 is further divided into three sections including a section A, a section B, and a section C. An energization pattern is set individually for each of the section A, the section B, and the section C. The period (electrical angle) of each of the section A, the section B, and the section C changes according to the value of the advance angle and the value of the energization angle.

(66) Since the switching elements **51a** to **51f** each repeat "ON" and "OFF" in the PWM-controlled state, the waveform is in fact a rectangular wave shape having projections and indentations. However, for convenience of illustration, in FIG. 4 and FIG. 6 (to be described later), the "ON/OFF" of each of the switching elements **51a** to **51f** is not specified, but is labeled as "PWM phase". Herein, in FIG. 4, the "PL" state is labeled as "PWM phase (MAX Duty)", the "PS" state is labeled as "PWM phase (MIN Duty)", and the "PM" state is labeled as "PWM phase (Duty=50)".

(67) For example, in the stage 1 defined between the Hall edge HE1 and the Hall edge HE2, the energization pattern of the section A is a combination of the states of the switching elements **51a** to **51f** being "PL", "PL", "PS", "PS", "PM", and "PM", respectively. Further, the energization pattern of the section B is a combination of the states of the switching elements **51a** to **51f** being "PL", "PL", "PS", "PS", "PL", and "PL", respectively. Further, the energization pattern of the section C is a combination of the states of the switching elements **51a** to **51f** being "PM", "PM", "PS", "PS", "PL", and "PL", respectively.

(68) Further, FIG. 5 is a table summarizing an example of the non-free energization patterns of the first method shown in FIG. 4. The ROM (not shown) in the control part **40** stores the non-free energization patterns in a format as shown in FIG. 5, for example. In FIG. 5, "PL" represents the fourth state in which PWM control is performed according to a PWM signal with the maximum duty ratio, "PS" represents the fifth state in which PWM control is performed according to a PWM signal with the minimum duty ratio, and "PM" represents the sixth state in which PWM control is performed according to a PWM signal with the intermediate duty ratio between the maximum duty ratio and the minimum duty ratio. Thus, in the non-free driving mode (first method), the driving control part **43** controls conduction (energization) of the switching elements **51a** to **51f** according to the non-free energization patterns above.

(69) <Second Method of Non-Free Energization>

(70) In the driving and energization based on the non-free energization pattern using the second method, the states (1) to (3) above are respectively changed to three states (7) to (9) below. In the state (1) and the state (7), since the control on each of the switching elements is the same, actually the state has not changed.

(71) (7) Seventh state: The first state is maintained, and the switching element **51a** to **51f** is continuously turned on ("ON").

(72) (8) Eighth state: The third state is changed to an eighth state (hereinafter referred to as a “P1” state) in which PWM control is performed according to a PWM signal with the instruction duty ratio inputted from outside.

(73) (9) Ninth state: The second state is changed to a ninth state (hereinafter referred to as a “P2” state) in which PWM control is performed according to a PWM signal with a duty ratio that is one half of the instruction duty ratio inputted from outside.

(74) In other words, the timings at which the U-phase, V-phase, and W-phase windings (21u, 21v, 21w) respectively become the P2 state (ninth state) in the non-free energization pattern are the same as the timings at which the U-phase, V-phase, and W-phase windings (21u, 21v, 21w) respectively become OFF-phase windings (second state) in the rectangular wave energization pattern.

(75) Thus, it is possible to address the problem that the control circuit (control part 40) that drives the switching elements malfunctions when a negative voltage is generated at an input terminal of the motor 2 upon switching of the energization pattern from the first state to the second state. Further, even at the timing (phase open period: period of being in the second state) of being free in 120-degree rectangular wave energization, PWM control is performed as in the ninth state, which results in 180-degree energization and a smooth current waveform at the time of commutation, so it becomes possible to expect an effect of quieting a driving sound (reducing motor operation sound). In other words, similar effects can be expected using the driving and energization method according to the non-free energization pattern of either the first method or the second method.

(76) Herein, in this embodiment, the duty ratio in the seventh state is 100%. For example, in the case where the instruction duty ratio is 80%, the duty ratio in the eighth state becomes 80%, and the duty ratio in the ninth state becomes 40% according to $80 \div 2$. The instruction duty ratio is assumed to be stored in advance to the ROM (not shown) of the control part 40 by the user.

(77) FIG. 6 is a diagram showing an example of the non-free energization patterns of the second method. FIG. 6 is a diagram showing the correspondence between the position detection signals (Hu, Hv, Hw) and the energization patterns of the respective switching elements 51a to 51f, with the horizontal axis representing the electrical angle.

(78) The example of energization control shown in FIG. 6 is the case where the advance angle is 20 degrees and the energization angle is 130 degrees. The energization pattern is a combination of any of states of the respective switching elements 51a to 51f, including a state (seventh state) of maintaining the continuously on state (“ON”), i.e., “ON” (first state), a state changed from being continuously turned off (“OFF”) (period other than “ON” or “PWM”), i.e., “OFF” (second state), to “P2” (ninth state), and a state changed from being controlled to be turned “on” or “off” at a specific cycle (PWM-controlled state) (“PWM”), i.e., “PWM” (third state), to “P1” (eighth state). Each of the stages 1 to 6 is further divided into three sections including a section A”, a section B”, and a section C”. An energization pattern is set individually for each of the section A”, the section B”, and the section C”. The period (electrical angle) of each of the section A”, the section B”, and the section C” changes according to the value of the advance angle and the value of the energization angle. Herein, in FIG. 6, the “P1” state is labeled as “PWM phase (Instruction Duty)”, and the “P2” state is labeled as “PWM phase (1/2 Duty)”.

(79) For example, in the stage 1 defined between the Hall edge HE1 and the Hall edge HE2, the energization pattern of the section A” is a combination of the states of the switching elements 51a to 51f being “1”, “0”, “P1”, “P1”, “P2”, and “P2”, respectively. Further, the energization pattern of the section B” is a combination of the states of the switching elements 51a to 51f being “1”, “0”, “P1”, “P1”, “1”, and “0”, respectively. Further, the energization pattern of the section C” is a combination of the states of the switching elements 51a to 51f being “P2”, “P2”, “P1”, “P1”, “1”, and “0”, respectively. Further, FIG. 7 is a table summarizing an example of the non-free energization patterns of the second method shown in FIG. 6. The ROM (not shown) in the control part 40 stores the non-free energization patterns in a format as shown in FIG. 7, for example. In

FIG. 7, “P1” represents the eighth state in which PWM control is performed according to a PWM signal with the instruction duty ratio inputted from outside, “P2” represents the ninth state in which PWM control is performed according to a PWM signal with a duty ratio that is one half of the instruction duty ratio, “1” represents “ON”, and “0” represents “OFF”. Thus, in the non-free driving mode (second method), the driving control part **43** controls conduction (energization) of the switching elements **51a** to **51f** according to the non-free energization patterns above.

(80) As described above, the rectangular wave driving mode is a driving mode of the rectangular wave driving method and is a driving mode of an output higher than the non-free driving mode. Further, the non-free driving mode is a driving mode of the non-free driving method which outputs a driving signal of an intermediate power by PWM control for an open period of a non-energized phase among driving signal lines of a plurality of phases driving the motor **2**, and the non-free driving mode is a driving mode of an output lower than the rectangular wave driving mode. Herein, the driving signal of the intermediate power is a PWM signal with an intermediate duty ratio between the PWM signal with the maximum duty ratio and the PWM signal with the minimum duty ratio (first method), or a PWM signal with a duty ratio that is one half of the instruction duty ratio inputted from outside (second method).

(81) Returning to the description of FIG. **1**, the driving control part **43** performs control of switching between the non-free driving mode and the rectangular wave driving mode based on the vehicle speed detected by the vehicle speed detection part **42**. The driving control part **43** switches from the non-free driving mode to the rectangular wave driving mode in the case where the vehicle speed becomes equal to or greater than a predetermined rectangular wave threshold V_{th1} (equal to or greater than a first threshold). Further, the driving control part **43** switches from the rectangular wave driving mode to the non-free driving mode in the case where the vehicle speed becomes equal to or less than a non-free threshold V_{th2} (equal to or less than a second threshold) lower than the rectangular wave threshold V_{th1} (first threshold). Herein, referring to FIG. **8**, the rectangular wave threshold V_{th1} and the non-free threshold V_{th2} will be described.

(82) FIG. **8** is a diagram showing an example of a switching operation of the driving mode of the motor control device **150** according to this embodiment. In FIG. **8**, the horizontal axis represents time, and the vertical axis represents the vehicle speed. Further, a waveform $W1$ shows an example of a change in the vehicle speed over time. Further, a period $TR1$ and a period $TR3$ represent periods of the non-free driving mode, and a period $TR2$ represents a period of the rectangular wave driving mode.

(83) As shown in FIG. **8**, when the vehicle speed increases and becomes equal to or greater than the rectangular wave threshold V_{th1} , the driving control part **43** switches from the non-free driving mode to the rectangular wave driving mode. Further, when the vehicle speed decreases and becomes equal to or less than the non-free threshold V_{th2} , the driving control part **43** switches from the rectangular wave driving mode to the non-free driving mode. Herein, the rectangular wave threshold V_{th1} is a value in a range of 1.1 times to 6.6 times the non-free threshold V_{th2} .

(84) Next, an operation of the motor control device **150** according to this embodiment will be described with reference to the drawings. FIG. **9** is a flowchart showing an example of the switching operation of the driving mode of the motor control device **150** according to this embodiment.

(85) As shown in FIG. **9**, first, the control part **40** of the motor control device **150** detects a vehicle speed (step **S101**). The vehicle speed detection part **42** of the control part **40** detects the vehicle speed based on a vehicle speed signal outputted from the ECU **4**.

(86) Next, the driving control part **43** of the control part **40** determines whether the driving mode of the motor **2** is the non-free driving mode (step **S102**). In the case where the driving mode is the non-free driving mode (step **S102**: YES), the driving control part **43** proceeds to step **S103**. In the case where the driving mode is not the non-free driving mode (step **S102**: NO), the driving control part **43** proceeds to step **S107**.

(87) In step **S103**, the driving control part **43** determines whether the vehicle speed is equal to or greater than a rectangular wave threshold **Vth1** (vehicle speed $\geq$ rectangular wave threshold **Vth1**). In the case where the vehicle speed is equal to or greater than the rectangular wave threshold **Vth1** (step **S103**: YES), the driving control part **43** proceeds to step **S104**. In the case where the vehicle speed is less than the rectangular wave threshold **Vth1** (step **S103**: NO), the driving control part **43** returns to step **S101**.

(88) In step **S104**, the driving control part **43** switches from the non-free driving mode to the rectangular wave driving mode. Next, the driving control part **43** initializes an accumulated point value (load integrated value) (step **S105**). The driving control part **43** initializes (resets to "0") the load integrated value generated by the load integrated value generation part **41**.

(89) Next, the driving control part **43** cancels an energy mode (step **S106**). Herein, the energy mode is a protection mode that restricts the output in the case where the load integrated value becomes very large (in the case where the load increases). After the process in step **S106**, the driving control part **43** returns to step **S101**.

(90) Further, in step **S107**, the driving control part **43** determines whether the vehicle speed is equal to or less than a non-free threshold **Vth2** (vehicle speed $\leq$ non-free threshold **Vth2**). In the case where the vehicle speed is equal to or less than the non-free threshold **Vth2** (step **S107**: YES), the driving control part **43** proceeds to step **S108**. In the case where the vehicle speed is greater than the non-free threshold **Vth2** (step **S107**: NO), the driving control part **43** returns to step **S101**.

(91) In step **S108**, the driving control part **43** switches from the rectangular wave driving mode to the non-free driving mode. Next, the driving control part **43** initializes an accumulated point value (load integrated value) (step **S109**). The driving control part **43** initializes (resets to "0") a load integrated value generated by the load integrated value generation part **41**.

(92) Next, the driving control part **43** cancels the energy mode (step **S110**). After the process in step **S110**, the driving control part **43** returns to step **S101**.

(93) Further, referring to FIG. **10**, a switching operation of the driving mode in the case where the vehicle speed signal cannot be detected will be described. FIG. **10** is a flowchart showing another example of the switching operation of the driving mode of the motor control device **150** according to this embodiment.

(94) As shown in FIG. **10**, first, the driving control part **43** of the motor control device **150** checks for an abnormality of a vehicle speed signal (step **S201**). Next, the driving control part **43** determines whether there is an abnormality of the vehicle speed signal (step **S202**). That is, the driving control part **43** determines whether the vehicle speed can be detected normally. In the case where there is an abnormality of the vehicle speed signal (the vehicle speed cannot be detected) (step **S202**: YES), the driving control part **43** proceeds to step **S203**. In the case where there is no abnormality of the vehicle speed signal (step **S202**: NO), the driving control part **43** proceeds to step **S204**.

(95) In step **S203**, the driving control part **43** changes to the rectangular wave driving mode as an emergency operation. After the process in step **S203**, the driving control part **43** returns to step **S201**.

(96) Further, in step **S204**, the driving control part **43** cancels the emergency operation. After the process in step **S204**, the driving control part **43** returns to step **S201**.

(97) Next, referring to FIG. **11**, a modification example of the switching operation of the driving mode of the motor control device **150** according to this embodiment will be described. FIG. **11** is a flowchart showing a modification example of the switching operation of the driving mode of the motor control device according to this embodiment. Herein, a modification example in which, of the thresholds of the vehicle speed described above, the rectangular wave threshold **Vth1** and the non-free threshold **Vth2** are equal values will be described.

(98) As shown in FIG. **11**, first, the control part **40** of the motor control device **150** detects a vehicle speed (step **S301**). The vehicle speed detection part **42** of the control part **40** detects the vehicle

speed based on a vehicle speed signal outputted from the ECU **4**.

(99) Next, the driving control part **43** of the control part **40** determines whether the vehicle speed is equal to or greater than a threshold V_{th3} (vehicle speed $\geq$ threshold V_{th3}) (step **S302**). In the case where the vehicle speed is equal to or greater than the threshold V_{th3} (step **S302**: YES), the driving control part **43** proceeds to step **S303**. In the case where the vehicle speed is less than the threshold V_{th3} (step **S302**: NO), the driving control part **43** proceeds to step **S304**.

(100) In step **S303**, the driving control part **43** changes to the rectangular wave driving mode. After the process in step **S303**, the driving control part **43** proceeds to step **S305**. Further, in step **S304**, the driving control part **43** changes to the non-free driving mode. After the process in step **S304**, the driving control part **43** proceeds to step **S305**.

(101) In step **S305**, the driving control part **43** determines whether the driving mode has been changed. In the case where the driving mode has been changed (step **S305**: YES), the driving control part **43** proceeds to step **S306**. In the case where the driving mode has not been changed (step **S305**: NO), the driving control part **43** returns to step **S301**.

(102) In step **S306**, the driving control part **43** initializes an accumulated point value (load integrated value). The driving control part **43** initializes (resets to "0") the load integrated value generated by the load integrated value generation part **41**.

(103) Next, the driving control part **43** cancels the energy mode (step **S307**). After the process in step **S307**, the driving control part **43** returns to step **S301**.

(104) As described above, the driving control part **43** may switch to the rectangular wave driving mode in the case where the vehicle speed is equal to or greater than the predetermined threshold V_{th3} , and may switch to the non-free driving mode in the case where the vehicle speed is less than the threshold V_{th3} (less than the predetermined threshold).

(105) Next, referring to FIG. **12**, an example of a case of applying the motor device **100** above to a wiper device will be described. FIG. **12** is a configuration view showing an example of a wiper device **200** according to this embodiment.

(106) As shown in FIG. **12**, the wiper device **200** performs a wiping operation on a window surface of a window glass **10** of a vehicle **1**. The wiper device **200** includes a motor device **100**, a link mechanism **11**, two wiper arms **12**, and wiper blades **13** mounted at tips of the respective wiper arms **12**.

(107) The motor device shown in FIG. **12** is the motor device **100** of this embodiment as described above, and detailed descriptions thereof will be omitted herein. The motor device **100** includes a motor **2** and a motor control device **150**.

(108) The wiper arm **12** operates on the window surface of the window glass **10** by the rotational driving of the motor device **100** and performs the wiping operation with the wiper blade **13** mounted at the tip. The two wiper arms **12** are connected by the link mechanism **11**.

(109) The wiper blade **13** is provided to be pressed against the window glass **10** by the wiper arm **12**. The wiper blade **13** includes a blade rubber (not shown) that is held by a blade holder mounted at the tip of the wiper arm **12**. Upon swinging of the wiper arm **12** by the motor device **100**, the wiper blade **13** reciprocates over a wiping range on the outer surface of the window glass **10** and wipes the window glass **10** with the blade rubber (not shown).

(110) As described above, a motor control device **150** according to this embodiment is a motor control device which controls a motor **2** having a rotor shaft **22a** (output shaft) and includes a vehicle speed detection part **42** and a driving control part **43**. The vehicle speed detection part **42** detects a vehicle speed, which is a traveling speed of a vehicle mounted with the motor **2**. Based on the vehicle speed detected by the vehicle speed detection part **42**, the driving control part **43** performs control of switching between a non-free driving mode (first driving mode) which drives the motor **2**, and a rectangular wave driving mode (second driving mode) which is a high-output driving mode in which a minimum rotational speed at which the output shaft is capable of being rotationally driven by an inputted power is higher than in the non-free driving mode (first driving

mode). Herein, the second driving mode is a high-output driving mode in which a number of revolutions and an output of the motor 2 are higher than in the first driving mode at the same load, power, and duty.

(111) In the motor control device 150 according to this embodiment, for example, in the case of being applied to a wiper device 200, there is a tendency for the load of the motor 2 to increase as the vehicle speed increases. Thus, the motor control device 150 according to this embodiment can appropriately perform motor driving even in the case where a high load is generated from an initial stage of driving. That is, in the motor control device 150 according to this embodiment, for example, the wiping operation can be performed without stop even in initial wiping immediately after wiper actuation during vehicle traveling.

(112) Further, since the motor control device 150 according to this embodiment can appropriately switch to the non-free driving mode (first driving mode) in the case where the vehicle is stopped, it is possible to quiet a driving sound (reduce motor operation sound).

(113) Further, in this embodiment, the driving control part 43 switches from the non-free driving mode (first driving mode) to the rectangular wave driving mode (second driving mode) in the case where the vehicle speed becomes equal to or greater than a predetermined rectangular wave threshold V_{th1} (equal to or greater than a first threshold). Further, the driving control part 43 switches from the rectangular wave driving mode (second driving mode) to the non-free driving mode (first driving mode) in the case where the vehicle speed becomes equal to or less than a non-free threshold V_{th2} (equal to or less than a second threshold) lower than the rectangular wave threshold V_{th1} .

(114) Accordingly, the motor control device 150 according to this embodiment switches to the rectangular wave driving mode (second driving mode) in the case where the vehicle speed is high (load is large), and switches to the non-free driving mode (first driving mode) in the case where the vehicle speed is low (load is small). Thus, it is possible to realize both reduction of the motor operation sound in the case of a small load and reliable motor driving in the case of a large load. Further, by using two thresholds including the rectangular wave threshold V_{th1} and the non-free threshold V_{th2} , the motor control device 150 according to this embodiment can reduce a malfunction such as frequent switching of the driving mode due to noise, for example.

(115) Further, in this embodiment, the rectangular wave threshold V_{th1} is a value in a range of 1.1 times to 6.6 times the non-free threshold V_{th2} . Thus, the motor control device 150 according to this embodiment can appropriately respond to even a vehicle such as an automobile traveling at a high speed of 200 km/h or more.

(116) Further, in this embodiment, the second driving mode is a driving mode of a rectangular wave driving method (rectangular wave driving mode), and the first driving mode is a driving mode of a non-free driving method (non-free driving mode). The driving mode of the non-free driving method (non-free driving mode) is a mode that outputs a driving signal of an intermediate power by PWM control for an open period (free period) of a non-energized phase among driving signal lines of a plurality of phases driving the motor 2. The driving control part 43 switches from the non-free driving mode to the rectangular wave driving mode in the case where the vehicle speed becomes equal to or greater than the rectangular wave threshold V_{th1} . Further, the driving control part 43 switches from the rectangular wave driving mode to the non-free driving mode in the case where the vehicle speed becomes equal to or less than the non-free threshold V_{th2} . Thus, the motor control device 150 according to this embodiment can appropriately switch the driving mode between the rectangular wave driving mode and the non-free driving mode according to the vehicle speed.

(117) Further, in this embodiment, the driving signal of the intermediate power is a PWM signal with an intermediate duty ratio between a PWM signal with a maximum duty ratio and a PWM signal with a minimum duty ratio (first method), or is a PWM signal with a duty ratio that is one half of an instruction duty ratio inputted from outside (second method). Thus, the motor control

device **150** according to this embodiment can appropriately perform motor driving by the non-free driving mode.

(118) Further, in this embodiment, the motor **2** is a brushless motor having windings (**21u**, **21v**, **21w**) of three phases. The driving control part **43** controls conduction of a plurality of switching elements **51a** to **51f** included in an inverter **50**, which generates driving signals of three phases, by switching between the non-free driving mode (first driving mode) and the rectangular wave driving mode (second driving mode). Thus, the motor control device **150** according to this embodiment can perform more appropriate motor driving on the brushless motor.

(119) Further, in this embodiment, the driving control part **43** may switch to the rectangular wave driving mode (second driving mode) in the case where the vehicle speed is equal to or greater than a predetermined threshold V_{th3} , and switch to the non-free driving mode (first driving mode) in the case where the vehicle speed is less than the threshold V_{th3} . Thus, the motor control device **150** according to this embodiment can realize both reduction in the motor operation sound in the case of a small load and reliable motor driving in the case of a large load.

(120) Further, in this embodiment, the driving control part **43** initializes (sets to "0") an accumulated point value (load integrated value) in the case where the operation mode is switched. Thus, the motor control device **150** according to this embodiment can reduce occurrence of an error in the accumulated point value (load integrated value) and occurrence of a malfunction when switching the driving mode.

(121) Further, in this embodiment, the driving control part **43** cancels an energy mode, which is a protection mode that restricts the output when the accumulated point value (load integrated value) becomes very large (when the load increases), in the case where the operation mode is switched. Thus, the motor control device **150** according to this embodiment can reduce occurrence of a malfunction due to the protection mode when switching the driving mode.

(122) Further, in this embodiment, the driving control part **43** switches to the rectangular wave driving mode (second driving mode) as an emergency operation in the case where the vehicle speed cannot be detected. Thus, the motor control device **150** according to this embodiment can reliably perform motor driving by switching to the rectangular wave driving mode (second driving mode) in the event of an abnormality.

(123) Further, a motor device **100** according to this embodiment includes the motor **2** and the motor control device **150** described above. Thus, the motor device **100** according to this embodiment achieves the same effects as the motor control device **150**, and can appropriately perform motor driving even in the case where a high load is generated from an initial stage of motor driving.

(124) Further, a wiper device **200** according to this embodiment includes the motor **2** and the motor control device **150** described above. The motor control device **150** causes a wiper member (wiper arm **12** and wiper blade **13**) to perform a wiping operation on a window surface using the motor **2**.

(125) Thus, the wiper device **200** according to this embodiment achieves the same effects as the motor device **100** described above, and can appropriately perform motor driving even in the case where a high load is generated from the initial stage of motor driving.

(126) Further, a motor control method according to this embodiment is a motor control method which controls a motor **2** having a rotor shaft **22a** and includes a vehicle speed detection step and a driving control step. In the vehicle speed detection step, a vehicle speed detection part **42** detects a vehicle speed, which is a traveling speed of a vehicle mounted with the motor **2**. In the driving control step, based on the vehicle speed detected in the vehicle speed detection step, a driving control part **43** performs control of switching between a non-free driving mode (first driving mode) which drives the motor **2**, and a rectangular wave driving mode (second driving mode). The rectangular wave driving mode (second driving mode) is a high-output driving mode in which a minimum rotational speed at which the rotor shaft **22a** is capable of being rotationally driven by an inputted power is higher than in the non-free driving mode (first driving mode).

(127) Thus, the motor control method according to this embodiment achieves the same effects as

the motor device **100** described above, and can appropriately perform motor driving even in the case where a high load is generated from the initial stage of motor driving.

(128) In addition, the disclosure is not limited to the embodiments above and may be modified within a scope without departing from the spirit of the disclosure. For example, the embodiments above have described an example in which the motor control device **150** switches the driving mode according to the vehicle speed. However, the disclosure is not limited thereto, and the vehicle speed and the accumulated point value (load integrated value) may be combined to switch the driving mode.

(129) Further, the embodiments above have described an example of switching between the non-free driving mode (first driving mode) and the rectangular wave driving mode (second driving mode), but the disclosure is not limited thereto, and other driving modes may also be switched.

(130) Further, the embodiments above have described an example in which the motor device **100** is used in the wiper device **200**, but the disclosure is not limited thereto, and the motor device **100** may also be used for other purposes.

(131) In addition, each configuration included in the motor device **100** described above has a computer system therein. Then, a program for realizing the function of each configuration included in the motor device **100** described above may be recorded on a computer-readable recording medium, and by loading and executing the program recorded on this recording medium into the computer system, the process of each configuration included in the motor device **100** described above may be performed. Herein, “loading and executing a program recorded on a recording medium into a computer system” includes installing the program into the computer system. The term “computer system” as used herein includes an OS and hardware such as peripheral devices. Further, the “computer system” may include a plurality of computer devices connected via a network including a communication line such as the Internet, a WAN, a LAN, and a dedicated line. The “computer-readable recording medium” refers to a portable medium such as a flexible disk, a magneto-optical disk, a ROM, and a CD-ROM, or a storage device such as a hard disk built in a computer system. As described above, the recording medium on which the program is stored may also be a non-transitory recording medium such as a CD-ROM.

(132) Further, a part or all of the above functions may be realized as an integrated circuit such as a large scale integration (LSI). Each of the above functions may be individually configured into a processor, or a part or all of them may be integrated and configured into a processor. Further, the method of configuration into an integrated circuit is not limited to the LSI, but may also be realized by a dedicated circuit or a general-purpose processor. Further, when an integrated circuit configuration technology which replaces the LSI appears due to advances in semiconductor technology, an integrated circuit based on this technology may be used.

Claims

1. A motor control device which controls a motor having an output shaft, the motor control device comprising: a processor configured to: detect a vehicle speed, which is a traveling speed of a vehicle mounted with the motor; based on the vehicle speed detected perform control of switching between a first driving mode which drives the motor, and a second driving mode in which a number of revolutions of the output shaft and an output of the motor are higher than in the first driving mode; switch from the first driving mode to the second driving mode in a case where the vehicle speed becomes equal to or greater than a predetermined first threshold; and switch from the second driving mode to the first driving mode in a case where the vehicle speed becomes equal to or less than a second threshold lower than the first threshold; wherein the second driving mode is a driving mode of a rectangular wave driving method, the first driving mode is a driving mode of a non-free driving method that outputs a driving signal of an intermediate power by pulse width modulation control for an open period of a non-energized phase among driving signal lines of a

plurality of phases driving the motor, the processor is configured to switch from the driving mode of the non-free driving method to the driving mode of the rectangular wave driving method in a case where the vehicle speed becomes equal to or greater than the first threshold, and the processor is configured to switch from the driving mode of the rectangular wave driving method to the driving mode of the non-free driving method in a case where the vehicle speed becomes equal to or less than the second threshold.

2. The motor control device according to claim 1, wherein the first threshold is a value in a range of 1.1 times to 6.6 times the second threshold.

3. The motor control device according to claim 1, wherein the driving signal of the intermediate power is a pulse width modulation signal with an intermediate duty ratio between a pulse width modulation signal with a maximum duty ratio and a pulse width modulation signal with a minimum duty ratio, or is a pulse width modulation signal with a duty ratio that is one half of an instruction duty ratio inputted from outside.

4. The motor control device according to claim 1, wherein the motor is a brushless motor having windings of three phases, and the processor is further configured to control conduction of a plurality of switching elements included in an inverter, which generates driving signals of three phases, by switching between the first driving mode and the second driving mode.

5. A motor device comprising: the motor control device according to claim 1; and the motor.

6. A wiper device comprising: the motor control device according to claim 1; and the motor, wherein the motor control device causes a wiper member to perform a wiping operation on a window surface using the motor.

7. A motor control method which controls a motor having an output shaft, the motor control method comprising: detecting a vehicle speed, which is a traveling speed of a vehicle mounted with the motor; based on the vehicle speed detected performing control of switching between a first driving mode which drives the motor, and a second driving mode in which a number of revolutions of the output shaft and an output of the motor are higher than in the first driving mode; switching from the first driving mode to the second driving mode in a case where the vehicle speed becomes equal to or greater than a predetermined first threshold, and switching from the second driving mode to the first driving mode in a case where the vehicle speed becomes equal to or less than a second threshold lower than the first threshold; wherein the second driving mode is a driving mode of a rectangular wave driving method, the first driving mode is a driving mode of a non-free driving method that outputs a driving signal of an intermediate power by pulse width modulation control for an open period of a non-energized phase among driving signal lines of a plurality of phases driving the motor, the processor is configured to switch from the driving mode of the non-free driving method to the driving mode of the rectangular wave driving method in a case where the vehicle speed becomes equal to or greater than the first threshold, and the processor is configured to switch from the driving mode of the rectangular wave driving method to the driving mode of the non-free driving method in a case where the vehicle speed becomes equal to or less than the second threshold.
